

GaN Building Block for a 1200 V, 150 kW Stacked Polyphase Bridge Inverter

Design procedure, gate drive, layout and the Rev A fabrication board

Enes Ayaz

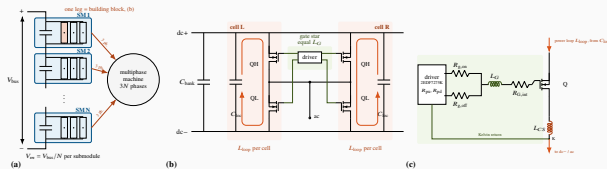
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KTH Royal Institute of Technology

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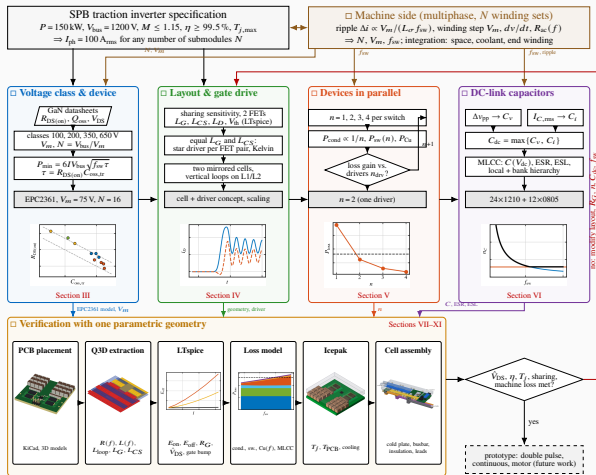
System and specification

Stacked polyphase bridge (SPB) inverter



- 1200 V bus, 150 kW traction inverter
- N three-phase submodules in series on the DC side, each feeding one winding set of a multiphase machine
- $V_m = V_{\text{bus}}/N$: the transistor voltage class becomes a **design variable**
- Phase current set by power and bus voltage: $\approx 100 \text{ A}_{\text{rms}}$ ($M = 1.15$)
- Chosen: $N = 16$, $V_m = 75 \text{ V}$, 3 kW per leg, 9 kW per submodule

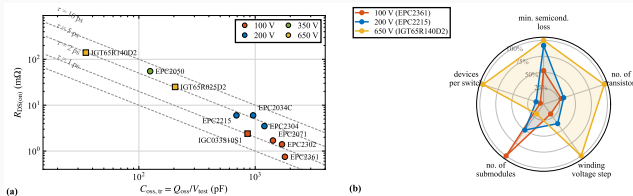
Design procedure



Specification → voltage class → layout /
 gate drive → number of parallel devices → DC link → verification, with the machine side feeding back
 N and f_{sw} .

Voltage class and transistor

Voltage class: why 100 V GaN



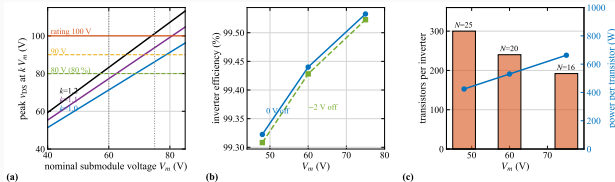
- At fixed bus voltage and phase current the minimum semiconductor loss scales with the device time constant $R_{DS(on)} C_{oss, tr}$
- 100 V GaN gives the lowest product among 100–650 V Si/SiC/GaN candidates
- Selected: **EPC2361**, 100 V, 0.75 m Ω typ, 3 $\times$ 5 mm, top-side cooling

EPC2361 key data

V_{DS} / transient	100 / 120 V
$R_{DS(on)}$	0.75 m Ω typ (1.0 max)
V_{GS} range	−4 ... +6 V
$V_{GS(th)}$	0.8 / 1.1 / 2.5 V
C_{iss} / C_{oss} / C_{rss}	3599 / 1019 / 13 pF
Q_G / Q_{GD} / Q_{OSS}	28 / 3.8 / 90 nC
$R_{G,int}$	0.4 Ω
$R_{\theta JC}$ top / $R_{\theta JB}$	0.2 / 1.5 K/W

- Low threshold and −4 V limit: gate drive needs care (bump, negative bias)
- Top of the package at **source potential**: thermal interface must insulate
- Seven interleaved stripes, pad 2 next to the gate is the natural Kelvin point

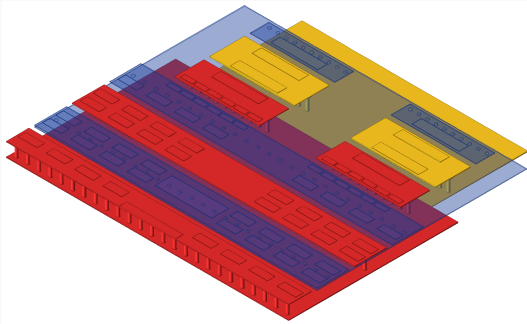
Voltage utilisation: 75 V or 60 V submodules



- 75 V nominal with $k = 1.2$
overshoot/ripple factor $\Rightarrow \approx 100$ V
peak at the device
- Keeping ≤ 80 V needs $V_m \leq 57$ V
($k = 1.2$) or ≤ 69 V ($k = 1$)
- Trade-off: 60 V submodules $\rightarrow$
 $N = 20$, more cells and drivers,
more margin
- Rev A board designed and tested
for 75 V

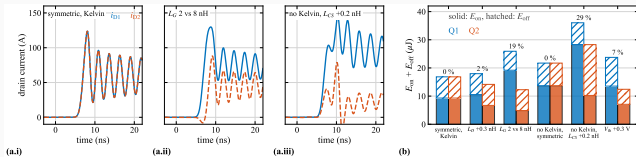
Layout and gate-drive concept

Building block layout: two mirrored cells, centre driver



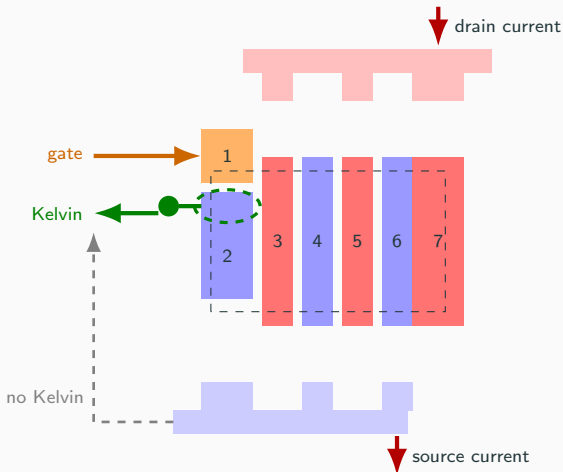
- One half-bridge leg, $2 \times$ EPC2361 per switch in **two mirrored commutation cells**
- Vertical power loop: L1 forward, L2 (DC-) return 0.1 mm below $\Rightarrow$ flux cancellation
- Local 0805 MLCCs at the drains, 1210 bank above both cells
- Driver on the symmetry line: identical gate loops to both transistors
- Q3D: $L_{\text{loop}} = 0.374 \text{ nH}$ per cell (copper + MLCC ESL)

Why symmetry: current-sharing sensitivity



- Static sharing is fine ($R_{DS(on)}$ has positive TC)
- Dynamic sharing is set by the gate circuits:
 - gate-loop mismatch 2 vs 8 nH
→ 19% imbalance, 4:1 E_{on}
 - common-source mismatch without Kelvin → 29%,
 $2.8 \times E_{on}$ in one device
- ⇒ star-connected gate drive with Kelvin returns, one driver channel per pair

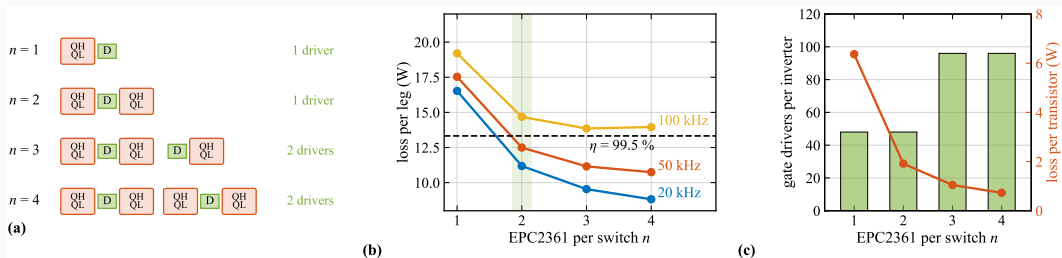
Kelvin source on the EPC2361



- Pad 2 is a separate source pad next to the gate pad; its gate end carries almost no load current
- Kelvin via at that end; return routed to the driver on its own path (L2 / L4)
- Gate loop (horizontal) $\perp$ power loop (vertical): minimum coupling
- Q3D: $L_{CS} = \mathbf{0.030\text{ nH}}$ with Kelvin, 0.061 nH without $\Rightarrow \approx 0.5$ vs 1.0 V subtracted from V_{GS} at 17 A/ns

Number of devices in parallel

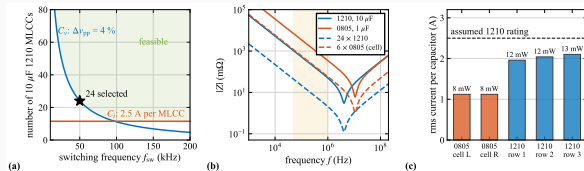
How many EPC2361 per switch?



- 1 → 2 devices: large drop in conduction loss and per-device loss
- 3 or 4 devices: only -1.3 / -1.7 W more, while output-charge loss grows and the drivers double (48 → 96)
- **Two per switch:** two cells with one centre driver

DC link

DC-link capacitors

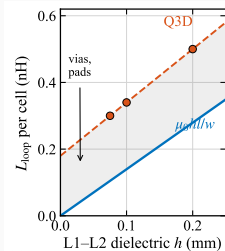


	local	bank
Part	TDK 0805	Murata 1210
	1 μF 100 V X7S	10 μF 100 V X7S
Number per leg	2×6	24
C_{eff} at 75 V	0.45 μF	3.0 μF
ESR / ESL	6 m Ω / 0.48 nH	3 m Ω / 0.5 nH
Ripple per part	1.1 A_{rms}	2.0–2.1 A_{rms}
Loss per part	7.5 mW	11.5–13.3 mW

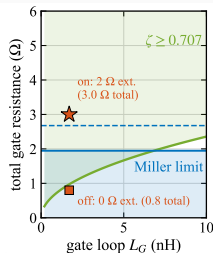
- Bank sized by voltage ripple below ≈ 50 kHz, by ripple current above
- Total capacitor loss ≈ 0.39 W per leg

**Parasitics, switching, losses,
thermal**

Extracted parasitics (Q3D)



(a)

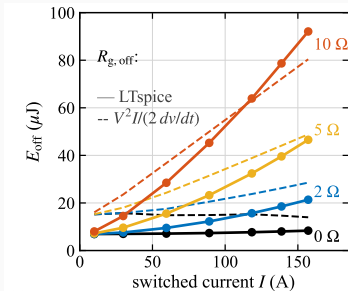


(b)

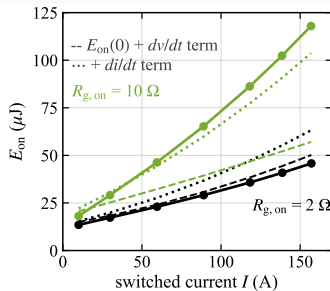
Power loop per cell (copper + ESL)	0.374 nH
Gate loop L_G per transistor	1.71 nH
L_{CS} , Kelvin / no Kelvin	0.030 / 0.061 nH
Gate loop – gate loop mutual	0.46 nH
Busbar, leg to leg (1 MHz)	3.35 nH
Busbar, series tab to leg	41 nH

Busbar: DC+ and DC– tabs leave at opposite ends (41 nH); bring them out together or use a laminated cell interconnect.

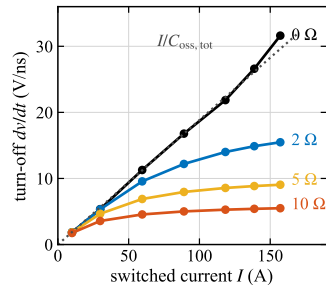
Switching behaviour



(a)



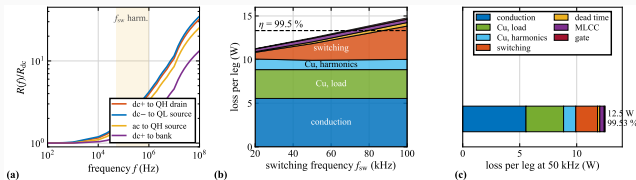
(b)



(c)

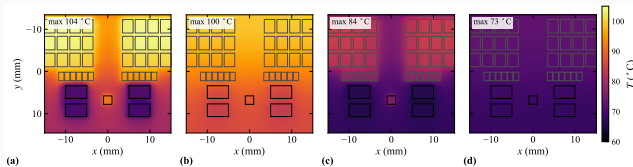
LTspice with the Q3D power loop and gate star, 75 V, $R_{g,\text{on}}/R_{g,\text{off}} = 2 / 0 \Omega$ external.

Losses and efficiency



- Rated point (75 V, 100 A_{rms}, 50 kHz): **12.5 W per leg**, $\eta = 99.53\%$
- Worst case 14.3 W (99.46 %)
- 1.91 W per transistor
- Copper and switching harmonics add $\approx 1/3$ to the DC copper loss

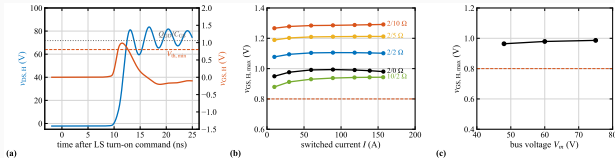
Thermal design



- Top-side cooling through an insulating interface to a liquid cold plate (integration concept)
- Icepak: $T_j \approx 64^\circ\text{C}$ at the rated point
- Lab: fan heatsink on the Rev A board (later slide)

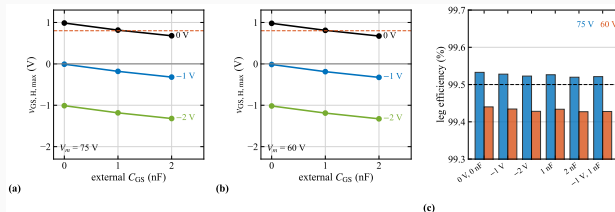
Gate bump and negative bias

Gate bump at the high side



- When the low side turns on, the high-side drain sees dv/dt : Miller charge flows into C_{GS}
- Bump $\approx Q_{GD}/C_{GS} \approx 1.06$ V, ≈ 2 ns, **independent of current and voltage**
- Below the 1.1 V typical threshold, close to the 0.8 V minimum: no shoot-through in simulation, but little margin

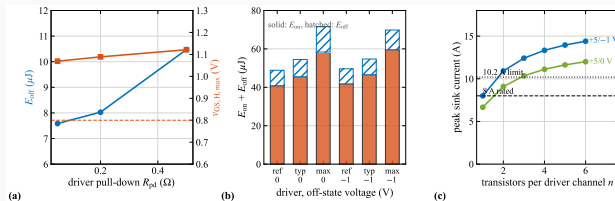
Mitigation: negative off-voltage



- With -1 V off-voltage the peak falls from 0.99 V to -0.01 V
- Undershoot -2.3 V, rating -4 V
- Cost: reverse conduction drop in the dead time rises by 1 V
- Dead-time loss $0.20 / 0.29 / 0.38$ W per leg for $0 / -1 / -2$ V
- **Chosen:** ≈ -1.24 V

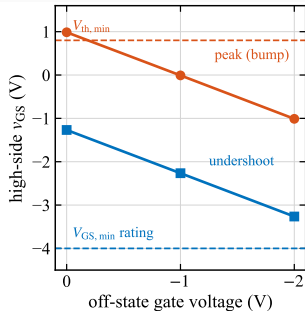
Gate-driver selection and scaling

Driver: Infineon 2EDF7275K

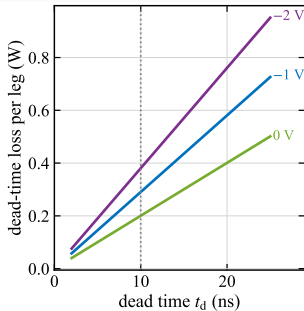


- Dual isolated channel, 4 A source / 8 A sink, 0.85 / 0.35 Ω
- 4 V UVLO: allows a split +5/ - 1.24 V supply per channel
- Sink strength sets the turn-off (E_{off}), not the bump
- LGA-13, 5 $\times$ 5 mm; functional isolation only

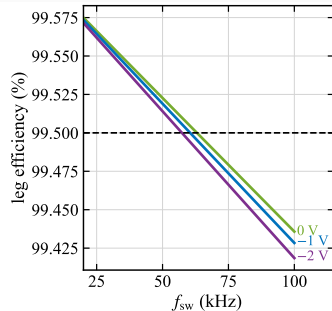
Negative rail vs dead-time loss



(a)

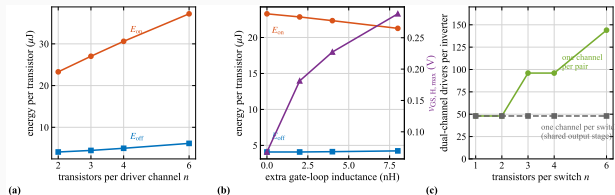


(b)



(c)

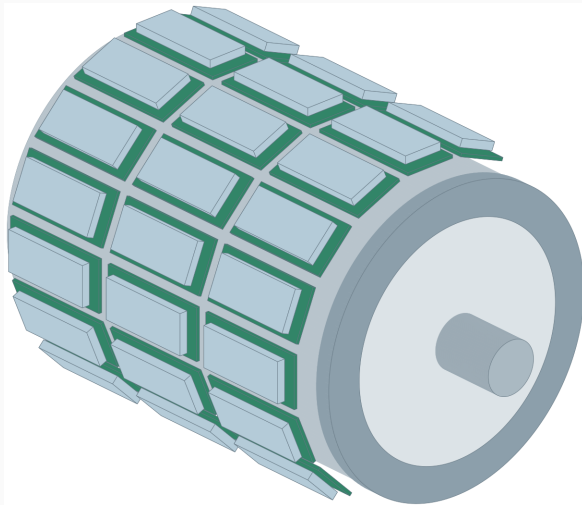
More parallel devices need more drivers



- One channel drives two transistors symmetrically
- Sharing one channel among 3 / 4 / 6 transistors: E_{on} per transistor +16 / +31 / +60 %
- Peak sink current 12–14 A, above the rating
- Longer gate tree brings the bump back
- $\Rightarrow$ scale in **pairs**, one channel per pair

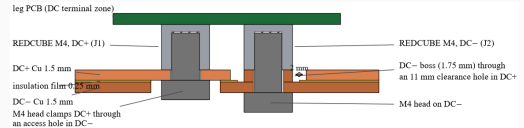
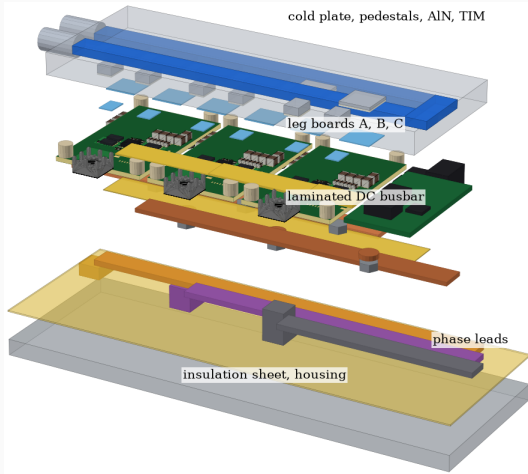
Integration and machine side

Machine-integrated drive concept



- One submodule = one cell assembly on a liquid cold plate (AlN insulation)
- Laminated busbar along the cells, phase leads to the winding sets
- Machine side feeds back into N (winding sets) and f_{sw} (ripple, iron loss)

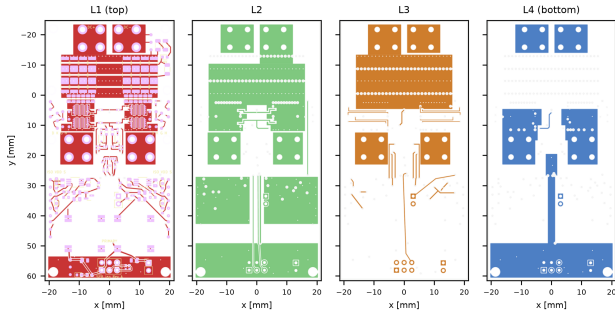
Cell assembly and busbar



- Leg-to-leg busbar inductance 3.35 nH (Q3D)
- Series tabs: 41 nH because DC+ and DC- leave at opposite ends

Rev A fabrication board

Rev A board: overview

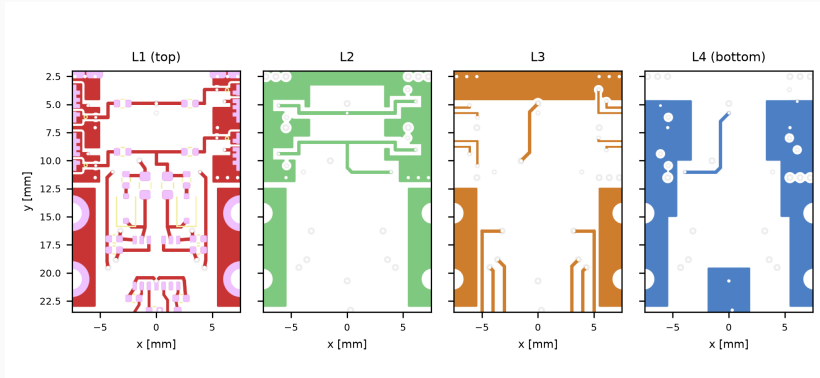


- 42×85.4 mm, 4 layers, 4×70 μ m
- DC terminals (top), two cells, driver corridor, AC terminals below each switch node, gate supplies, primary side (bottom)
- Outer strips: heatsink holes, probe landings, LEDs
- Fully scripted (KiCad 7 Python), DRC 0 / 0, wired schematic identical netlist

Stack-up and design rules

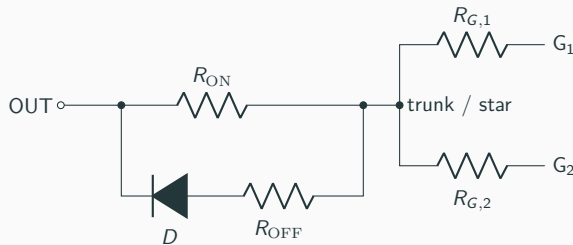
L1 70 μm prepreg 0.1 mm	power, pads, gate stars	all components vertical commutation loop
L2 70 μm core ≈ 1.1 mm	DC– plane	loop return, Kelvin collectors in narrow keepouts
L3 70 μm prepreg 0.1 mm	DC+ plane	HS gate trunk, supply columns, probe sense pairs HS gate trunk over its Kelvin return
L4 70 μm	AC plane	switch node to J3/J4, HS Kelvin return, primary ground
Rules	0.15/0.15 mm	power gaps 0.2 mm under solder mask (IPC-2221B B4: 0.13 mm for 51–100 V)
Vias	0.6/0.3 mm, 0.45/0.25 mm	no via-in-pad, standard process

Gate loops in the driver corridor



- HS: trunk on L3, Kelvin return directly below on L4; LS: star on L1, Kelvin collector on L2
- Kelvin collectors join on the symmetry line $\Rightarrow$ identical gate loops for both transistors

Separate turn-on and turn-off paths



$$R_{on,eff} = 2R_{ON} + R_{G,i}$$

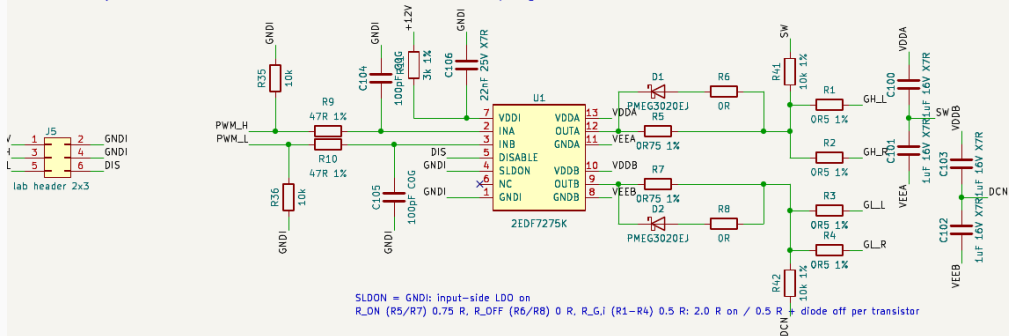
$$R_{off,eff} \approx 2(R_{ON} \parallel R_{OFF}) + R_{G,i} (+V_F)$$

Part	default	per transistor
R_{ON} (0603)	0.75Ω	on: 2.0Ω
R_{OFF} (0402)	0Ω	off: $0.5 \Omega + D$
D PMEG3020EJ	SOD323F	
$R_{G,i}$ (0402)	0.5Ω	

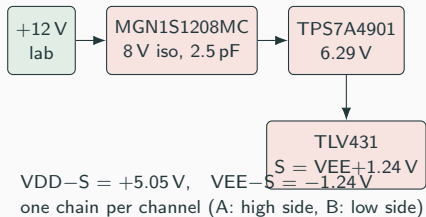
- Peak gate current 3.1 A (on) / 6.5 A (off): within 4 / 8 A
- Same as the simulated 2Ω / 0Ω external
- Values changeable by population
- 10 k Ω pull-down star $\rightarrow$ Kelvin per switch

Driver schematic (KiCad)

ers, VDDI, separate turn-on / turn-off networks, local decoupling



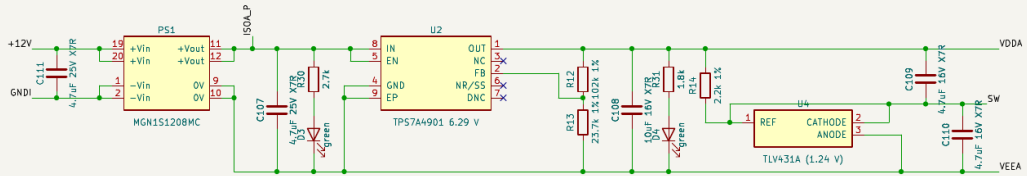
Isolated gate supplies



- Two isolated supplies: HS reference = switch node, LS reference = DC– (floats up to 1.2 kV in the stack)
- TPS7A4901 replaces LP2951: LP2951 has no VSSOP and its legacy chip needs $ESR \geq 30\text{ m}\Omega$
- $V_{OUT} = 1.185\text{ V} (1 + 102\text{ k}/23.7\text{ k})$; stable with ceramics $\geq 2.2\text{ }\mu\text{F}$
- TLV431 stable: $> 5\text{ }\mu\text{F}$ on S–VEE (unstable band 0.02–0.8 μF)
- Budget per channel at 50 kHz: 65 nC, 7.6 mA, 61 mW

Supply schematic (KiCad)

GATE SUPPLY A: MGN1S1208MC 12 V $\rightarrow$ 8 V iso, TPS7A4901 6.29 V, TLV431 sets $S = VEE + 1.24$ V ($S = SW$ Kelvin)



Isolation domains

Domains	outer	inner
DC+ – switch node	0.20	0.65
DC+ – DC–	0.70	1.07
HS – LS	0.20	0.40
HS – primary	1.37	1.08
LS – primary	1.50	0.75

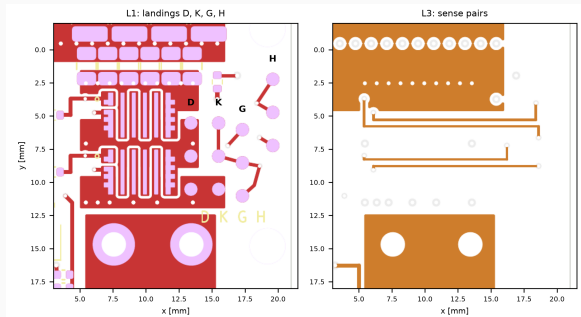
spacing [mm] on the routed board

minimum copper

- Power domains ≤ 100 V: 0.2 mm under mask
- Primary side separated functionally (lab board)
- In the 1200 V stack: PWM via fibre / reinforced isolator and isolated 12 V at the controller side; 2EDF7275K is functional only

Measurements and bring-up

Double-pulse test: measurement landings



- Probe pads only, outside the commutation loop; spring ground at 2.5 and 5.0 mm
- **D**: V_{DS} of Q3 / Q4 at the package
- **K**: v_{KS} = Kelvin vs power-source copper
- **G**: V_{GS} of Q3 / Q4 (sense pairs on L3)
- **H**: V_{GS} of Q1 / Q2 (isolated probe)
- Inductor current: Rogowski / Pearson off the board

E_{on} , E_{off} and per-transistor current

- $E = \int v_{DS} i_D dt$; E_{off} with $i_D = I_L$; E_{on} with I_L plus the high-side output charge $Q_{\text{OSS}} V_{\text{DC}}$
- Deskew < 50 ps: 100 ps at 50 V/ns and 100 A $\approx 0.75 \mu\text{J}$ error ($\approx 8\%$ of E_{on})
- Per-transistor current without a shunt (no extra loop inductance):

$$v_{KS} = L_s \frac{di_D}{dt} + R_s i_D \quad \Rightarrow \quad \Delta i_D(t) \approx \frac{1}{L_s} \int v_{KS} dt$$

$L_s \approx 0.03$ nH: ≈ 0.5 V at 17 A/ns; scale with $i_3 + i_4 = I_L$, ratio gives the dynamic sharing

- Same principle as stray-inductance di/dt sensing in SiC short-circuit protection (Sun, CPES 2019; Xue et al., TPEL 2020)
- Rogowski holes in the cells: not possible without opening the L1/L2 loop; coil bandwidth (20–50 MHz) too low for 2–3 ns edges

Bring-up: test points, LEDs, protection

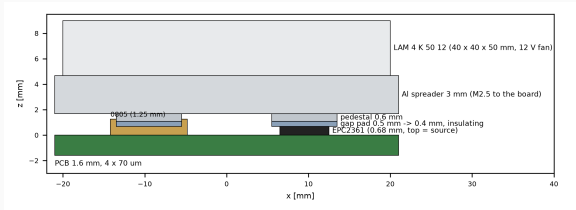
Item	nets	expected
TP15, D7	+12 V	12 V present
TP9/12, D3/5	ISO / VEE	MGN1 output 8–10 V
TP10/13, D4/6	VDD / VEE	6.29 V
TP11/14	S / VEE	1.24 V
TP16, D8 (red)	DC link	charged; 100 k bleeder
R35/R36	PWM	10 k pull-downs
R41/R42	star-Kelvin	10 k gate pull-downs
J7	+12 V	fan

Power-up order

1. 12 V (current limit), LEDs on, TP values
2. PWM without DC link: driver outputs on R5/R7, gates on R1–R4
3. DC link 12 → 24 → 48 → 75 V, double-pulse test
4. Heatsink, continuous operation

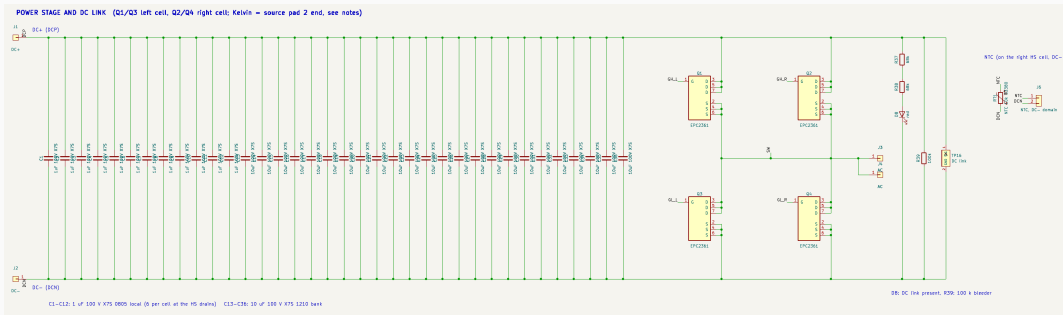
Power down in reverse: DC link first, then 12 V.

Cooling for the load test



- Fischer LAM 4 K 50 12 (0.65 K/W, 12 V fan from J7), off the shelf
- Al spreader $42 \times 14 \times 3$ mm (workshop part), $4 \times$ M2.5
- Insulating gap pad 0.5 mm, ≥ 5 W/mK: 5.3 K/W per transistor (dominant)
- $T_j \approx 57^\circ\text{C}$ (40°C ambient); flat plate + 1 mm pad: $\approx 70^\circ\text{C}$
- Next step: AlN insulator (≈ 1 K/W per transistor)

Power-stage schematic (KiCad)



Status and next steps

Done	DRC 0 violations / 0 unconnected; wired schematic = PCB netlist (37 nets); 3D models for all parts, STEP export; Gerber, drill, pick-and-place, BOM (35 lines, 107 parts + mechanics); datasheet checks: TPS7A4901 pins/stability, TLV431 pinout and stability, PMEG3020EJ
Prepared	Q3D re-extraction of the routed copper (<code>run_bb_q3d_fab.cmd</code>); needs the KTH licence (campus / VPN)
Open	Q3D result vs 0.374 nH design value; gap-pad part and spreader drawing; fab-house review of 0.2 mm gaps; isolation concept for the 1200 V stack

Next steps

1. Re-extract the routed board in Q3D, update loop and gate-loop values
2. Order Rev A, assemble, bring-up (12 V, PWM, then DC link)
3. Double-pulse test at 12–75 V: E_{on} , E_{off} , gate bump, sharing (D, G, K, H landings)
4. Continuous operation with the fan heatsink; thermal camera
5. Compare with LTspice / Q3D; decide 75 V vs 60 V submodules
6. Rev B: cell interconnect with laminated tabs, AlN interface, integration form factor (36 mm pitch)

Thank you

Questions?